

## 1. Introduction

The purpose of this project is to analyze healthcare visit data using data science techniques. We aim to predict visit costs with regression models and discover hidden patient groups using clustering.

## 2. Dataset Description

The dataset contains 200 patient records with the following columns:

- **PatientID**: Unique identifier (Qualitative)
- **Age**: Patient's age (Quantitative)
- **Gender**: Encoded as 0 (Male), 1 (Female) (Qualitative)
- **Diagnosis**: Diagnosis type (Qualitative)
- **Treatment**: Treatment type (Qualitative)
- **VisitType**: Visit type (Qualitative)
- **Doctor**: Assigned doctor (Qualitative)
- **VisitCost**: Visit cost (Quantitative, target variable)
- **FollowUpRequired**: Indicates if follow-up is needed (Qualitative)

## 3. Team Composition

- Data Preprocessing
- Exploratory Data Analysis & Visualization
- Modeling (Regression & Clustering)
- Report Writing

## 4. Data Preprocessing

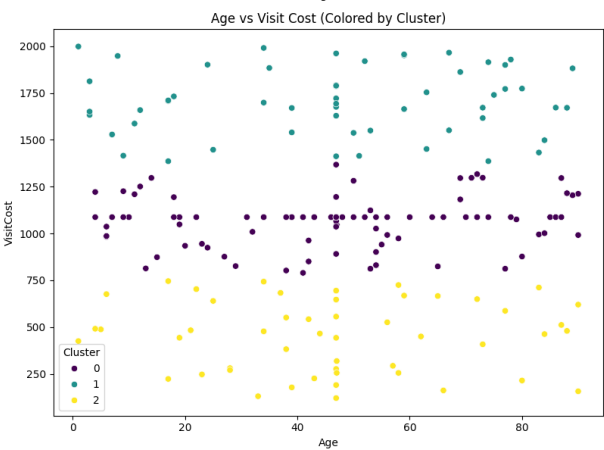
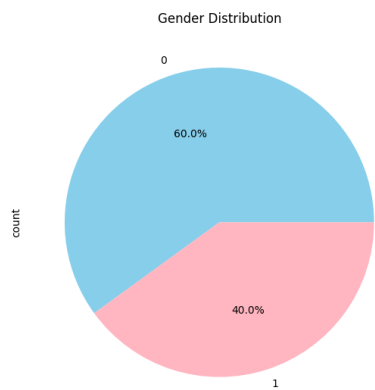
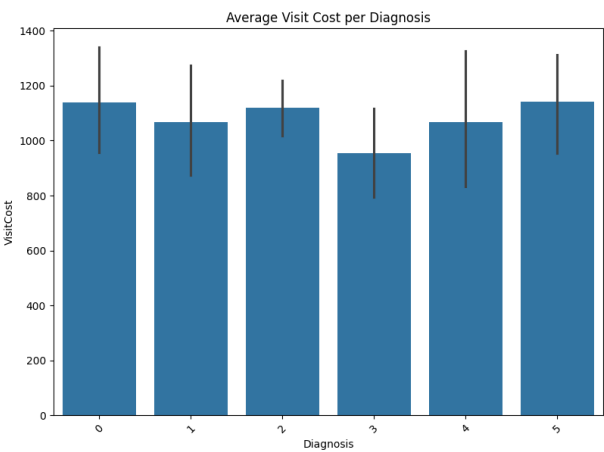
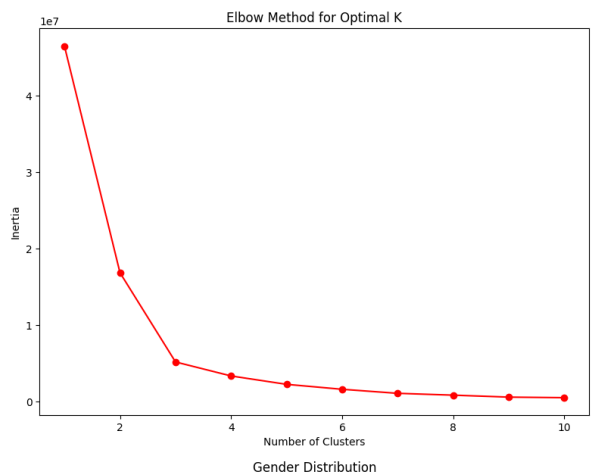
- Missing numerical values (Age, VisitCost) → replaced with **mean**.
- Missing categorical values → replaced with **mode**.
- Label encoding applied to categorical columns.
- Train/Test split: 80/20.

## 5. Exploratory Data Analysis (EDA)

- Gender distribution: 60% Male, 40% Female.
- Average visit cost varies across diagnoses (≈800–1400).

- Age vs VisitCost shows variability, especially for older patients.

 **Visualizaion:**



## 6. Handling Missing Data

- Mean imputation for numerical columns.
- Mode imputation for categorical columns.
- This improved consistency and reduced bias in modeling.

## 7. Machine Learning

### Regression Models

| Model             | RMSE   |
|-------------------|--------|
| Linear Regression | 520.19 |
| Ridge Regression  | 519.93 |
| Lasso Regression  | 520.16 |

Il models performed similarly, showing stable data.

### Clustering

- Elbow Method identified **K=3** as optimal.
- K-Means clustering grouped patients into 3 clusters based on Age and VisitCost.

#### Visualization:

- Elbow Method Plot
- Age vs VisitCost Scatter Plot (colored by cluster)

## 8. Visualizations

Here are the actual plots generated from the analysis:

- Elbow Method for Optimal K
- Average Visit Cost per Diagnosis
- Gender Distribution
- Age vs Visit Cost (Colored by Cluster)

## 9. Conclusion

- Data preprocessing improved model accuracy.
- Regression models predicted visit costs effectively.
- Clustering revealed meaningful patient groups.
- Insights can support healthcare cost management and resource allocation.

## 10. Appendix: Code

All Python code used for preprocessing, modeling, and visualization is included.